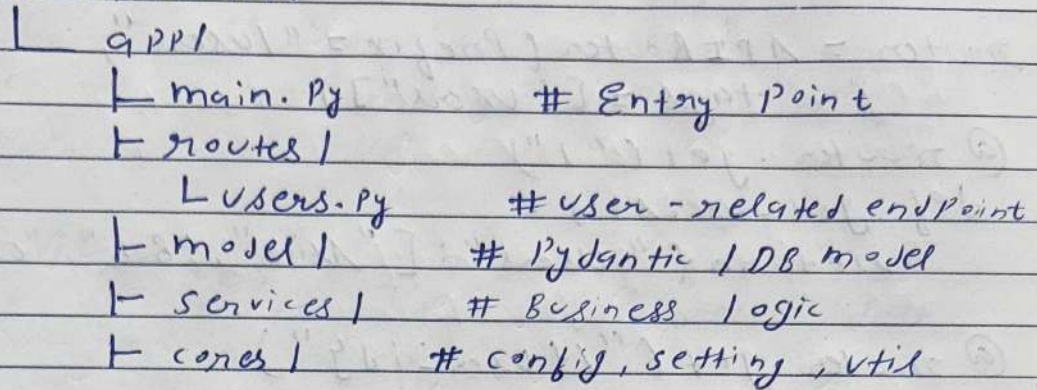




① FASTAPI Project Structure

Fastapi - app1



Analogy :

main.py = Main gate of Factory
routes/ = Different Department
services/ = Worker doing real work
models/ = Blueprint of data
core/ = Electricity, Water, rules

② Write Main.py (APP Entry point)

```
from fastapi import FastAPI
from app.routes import users
```

```
app = FastAPI()
app.include_router(users.router)
@app.get("/")
def root():
    return {"message": "API is running"}
```

③ Write Users Routes



From FastAPI import FastAPI, APIRouter

```
router = APIRouter(prefix="/users",  
                    tags=["users"])
```

```
@router.get("/")
```

```
def get_users():
```

```
    return {"users": ["Alice", "Bob", "Charlie"]}
```

```
@router.get("/{user-id}")
```

```
def get_users(user_id: int):
```

```
    return {"user-id": user_id, "name":  
            "Demo User"}
```

⑤ What is Routing?

Routing means:

Mapping URL + HTTP Method → to a
specific Python function

When a user send a request to your
server, routing decides which function
should run to handle that request.

Example

```
@router.get("/users/5")
```

```
def example():
```

```
    return {"message": "user 5"}
```

This means:

- ① If someone send a GET request.
- ② To the URL: /users/5
- ③ then the function example() will
run.



Breaking it Down

@router.get("/users/5")

- router → Handle routes
- get → HTTP method (get request)
- "/users/5" → URL path

""]

Analogy

Think of routing like a reception desk in an office.

- Some says: "I need billing department"
- Receptionist direct them to the correct room.

Similarly,

- client send: GET /users/5
- Router direct it to: example() function

⑥ Path Parameter

```
@router.get("/users/{user-id}")
def get-user(user-id: int):
    return {"user-id": user-id}
```

URL Example - /users/10

- {user-id} is a Path Parameter
- When someone visit:

GET /users/10

The value:

user-id = 10

get Passed into the function:

```
def get-user(user-id: int)
```

So Function receives:

user-id = 10



and returns:
{"user-id": 10}

Why is it called Path Parameter?

Because the value 10 is part of URL Path.

It directly identifies which resource you want.

Why do we use Path Parameters?

Use them when you want to:

- Get a specific user → /users/10
- " " Product → /product/5
- order → /order/1001
- blog post → /post/77

They are mainly used for:

IDs

Specific item

Required value

Identifying a resource

⑦ Query Parameters

```
@router.get("/users")
def get-users(limit: int = 10, skip:
               int = 0):
    return {"limit": limit, "skip": skip}
```

URL Example

/users?limit = 5 & skip = 10



What is happening here?

- ⊙ limit and skip are query parameters.
- ⊙ They come after ? in the URL.
- ⊙ They are written as:

?key = value

- multiple query parameters are separated with & :

?limit = 5 & skip = 10

When someone calls:

GET /users ? limit = 5 & skip = 10

The Function receive:

limit = 5

skip = 10

And return:

{

"limit": 5,

"skip": 10

}

What do limit and skip mean?

These are commonly used for pagination.

- ⊙ limit = 5 → Return only 5 users
- ⊙ skip = 10 → skip the first 10 users

So this request mean:

Skip the first 10 users and give me the next 5 users.



When do we use Query Parameter?

Query Parameter are used for:

- ① Filtering
- ② Pagination
- ③ Searching
- ④ Sorting
- ⑤ Optional setting

Examples:

1 Users? active = true

1 Products? category = electronics

1 orders? status = Shipped

1 users? search = join

Real world Analogy

① Path Param = House Number
"Deliver to House 12"

② Query Param = Delivery instruction
"Delivery before 5 PM, call first"